Vaggu Adarsh

ce23btech11056@iith.ac.in | adarshvaggu@gmail.com | +91 9182864943 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.75
Sri Chaitanya Vidya Bhavan	Class XII (Telangana State Board of Intermediate Education)	2023	96.70%
Sri Chaitanya , Nacharam	Class X (Board of Secondary Education, Telangana)	2021	10.00%

WORK EXPERIENCE

Research Assistant -- Vision-Language Model Evaluation | Aug 2026 - Present

- **Evaluated five small Vision-Language Models (VLMs)** across 385 responses on satellite precipitation-map interpretation using Qwen2.5-VL, SmolVLM, and LLaVA models.
- **Developed FIGScore**, a Format-Invariant Geospatial Similarity Score that evaluates factual agreement in meteorological information independent of wording and answer format.
- **Achieved a 14× correct-to-wrong score ratio with 0.0 format drift**, demonstrating stronger factual discrimination than existing similarity metrics; submitted the work to **TCCML @ NeurIPS 2026**.

POSITIONS OF RESPONSIBILITY

Finance Team Member -- Sunshine Club, IIT Hyderabad | 2025-2026

- Maintained and updated **financial expenditure records in Excel**, tracking bills and associated costs for the club.
- Prepared and submitted **advance forms, balance sheets, and other financial documentation** while keeping records accurate.
- Tracked and organized **club expenses and bill details**, helping maintain clear and up-to-date financial records.

Events Coordinator -- MILAN | 2025-2026

- Coordinated **event planning and on-ground execution** for MILAN, managing logistics and participant requirements.
- Worked closely with the **organizing team** to coordinate requirements and ensure timely execution of events.
- Supported **participant coordination and event logistics during execution**, helping activities run smoothly as scheduled.

Event Coordinator -- DIESTA | 2025-2026

- Assisted in **planning and coordinating event activities**, working with the organizing team on event requirements.
- Managed **on-ground operations and participant coordination** during scheduled activities.
- Supported the team in handling **event requirements and execution-related tasks**, helping ensure smooth conduct of activities.

EXTRACURRICULAR

Music, Paint, Art and Dance: Completed the course under **Oopali Operjitha**, gaining exposure to **ancient Indian paintings, traditional dance forms, music, and the cultural connection between different art forms**, with an interest in understanding their history.

Won 1st place in Clash Royale, Call of Duty, and Free Fire at the **MILAN College Tournament**.

Won Gold Medal the Civil Department Cricket Tournament as part of the Civil Day Sports Event.

Regularly participate in Badminton, Cricket, 8 Ball Pool, and Chess, maintaining active involvement in sports and

recreational activities.

Really enjoy playing Foosball, Billiards, and Table Tennis, and can spend hours playing them whenever I get the opportunity.

Sports: Enjoy playing **Football, Cricket, and Badminton**, regularly taking part in games and college activities, and have a strong interest in both team sports and individual competition.

SKILLS

Languages: Python, C++, SQL,C, JavaScript.

AI/ML: Machine Learning, Deep Learning, Transformers, LLMs, RAG, Embeddings.

GenAI: LangChain, LangGraph, Ollama, ChromaDB.

Development & Tools: Flask, REST APIs, Git, Docker.

Web: React.js, Node.js, Express.js, Flask, REST APIs, HTML, CSS.

Data Structures & Algorithms: Arrays, Strings, Linked Lists, Stacks & Queues, Trees, Graphs, Binary Search, Greedy, Dynamic Programming, Hashing, Sliding Window, Two Pointers

RELEVANT COURSES

Mathematics & Statistics: Calculus, Elementary Linear Algebra, Differential Equations, Introduction to Probability, Introduction to Statistics, Complex Variables, and Transform Techniques.

AI & Computing: Introduction to Programming and Artificial Intelligence(College). Machine Learning , Deep Learning , computer vision , NLP(NxtWave).

PROJECTS

AI Interview Assistant

- **Developed** an AI-powered voice interview platform using **Gemini**, with adaptive technical interviews and dynamically generated follow-up questions based on candidate responses.
- **Integrated** speech-to-text, streaming text-to-speech, automated **1-5 interview scoring**, and personalized feedback for an end-to-end interview experience.
- **Built** interview workflows using **LangChain and LangGraph**, connecting the frontend, backend, and AI services through REST APIs.
- **Tech Stack:** React.js, Flask, Python, Gemini , LangChain, LangGraph, AssemblyAI, Murf AI, REST APIs.

MODIS Land Surface Temperature Gap Filling

- **Developed** a Random Forest-based framework to reconstruct missing MODIS Land Surface Temperature (LST) data using temporal, spatial, and environmental features, combining temporal smoothing, DOY climatology, spatial interpolation, and Random Forest prediction for **10%, 20%, and 30% gaps**.
- **Engineered a preprocessing and modeling pipeline** using **17 temporal, spatial, and environmental features** to support large-scale LST reconstruction.
- **Achieved 46.4% RMSE improvement** over the interpolation baseline at 10% missing data, with the Random Forest model achieving **$R^2 = 0.891$** and maintaining strong reconstruction performance as the missing-data level increased.
- **Tech Stack:** Python, NumPy, Pandas, Scikit-learn, Random Forest, SciPy, Joblib, MODIS, SRTM, NDVI

NxtWatch

- **Developed** a responsive YouTube-like video streaming application with **Home, Trending, Gaming, Saved Videos, and Video Details** sections, supporting search, filtering, playback, and user interactions.
- **Implemented JWT authentication, protected routes, REST API integration, and React Context** for managing user state and saved videos, along with reusable components and responsive UI.
- **Added** dark theme support, loading/error states, and reusable UI components to improve the overall user experience.
- **Tech Stack:** React.js, JavaScript, React Router, React Context API, Styled-components, REST APIs, JWT, React Player.